Hip Implant Design Project Report

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Fahd A. KHALID-SALAKO [00036465]

Sabanci University Faculty of Engineering and Natural Sciences

Abstract

This project presents a patient-specific approach to hip implant design, addressing critical issues reported in literature, bothering hip prostheses such as stress shielding, poor osseointegration, and anatomical mismatch. In this project, we utilized diagnostic imaging obtained from the patient to generate 3D bone segments, from which a custom hip prosthesis is developed using Rhinoceros 3D CAD tools. The design process incorporates precise anatomical contouring of the femoral stem via lofted cross-sections, optimizing both fit and biomechanical performance. To enhance biological integration and load transfer, the femoral stem is engineered with a graded porosity profile, decreasing from 70% at the proximal end to 40% distally, in line with established biomechanical principles as cited. Additionally, elliptical support notches with 50% uniform porosity are integrated at high-stress regions to reinforce structural integrity and encourage directional bone ingrowth. The porosity gradient is algorithmically implemented using a custom Rhinoscript code, enabling user-defined parameters for tailored implant development. Additive manufacturing feasibility is demonstrated through structural continuity and no geometric artifacts. While the design meets immediate mechanical and anatomical criteria, future directions include finite element analyses, fatigue testing, and other optimization strategies to enhance mechanical efficiency and long-term viability of the implant.

Introduction/Literature Review

Hip replacement surgery is a surgical procedure conducted to restore functionality and resolve structural defects in the hip joint and proximal femoral anatomies. Conditions warranting hip replacement surgeries include osteoarthritis, fractured femoral necks, avascular necrosis, dysplasia, and inflammatory arthritis.[1] Biological and mechanical risk components including genetic and environmental factors, as well as patient-specific circumstances such as age, sex, trauma, and joint morphology contribute to a need for hip replacement in a wide sociodemographic range of patients globally.[2] Structurally, an increasingly prevalent indication for hip replacement surgery especially in younger patients is femoracetabular impingement.[3] Femoracetabular impingement is a condition with irregular bone growth within the hip joint, leading to abnormal contact between the femur and acetabulum and abrasion during movement. This can cause pain, limited range of motion, and may result in labral tears and damage to the articular cartilage, warranting either partial replacement or full replacement, depending on the extent of structural damage.

Research on hip implants/prostheses have demonstrated immense usefulness over the years and have allowed patients regain mobility and live relatively pain-free. However, two major issues continue to plague hip prostheses including stress-shielding and osseointegration, especially with the femoral stem.[4] The femoral stem is typically inserted into the patient's femoral cortical bone and connected with the acetabulum to facilitate transfer of mechanical stress from the hip to the healthy distal femur. However, design and osseointegration drawbacks cause stress-shielding and bone resorption over time, further leading to loosening of the femoral stem.[4] In several cases, patients require follow-up surgical procedures to retrieve the implanted prosthesis and replace it with new, bigger, femoral stem implant to fit the resorbed bone tissue area, exposing patients to potentially fatal complications.[5] Consequently, it is important to ensure that the femoral stem of a hip prosthesis integrates optimally with the bone and prevents stress-shielding, while maintaining mechanical stability and rigidity as required. Generally, hip implants are required, as proposed by Swansson,[6] to have some key properties, including:

1. Ensure joint clearance
2. Allow joint movement and stability
3. Have a simple and practical design
4. Provide long-term fixation
5. Resist degeneration
6. Be biocompatible and fit well with the patient's anatomy and biomechanical mechanism
7. Have a straightforward manufacturing process, simple manual operation tools and ensure the prosthetic stem is properly installed in place
8. Facilitate rehabilitation exercises.

Current approaches to design implants that meet these criteria include attempts to optimize femoral stem lengths; stem shape and geometry; porosity and functionally gradient constructs.[4] Nonetheless, the wide variety of patient anatomies and the ramifications of patient-specific

features warrant that the current prototypical approach to hip prosthesis design be revisited to prioritize patient-centered engineering and design strategies.

Therefore, for the purpose of this project, patient's 3D bone segments will be obtained and utilized to custom-design a hip prosthesis, with a focus on optimizing femoral stem structural design for improved outcomes, based on literature findings.

Solution Methodology

The custom hip prosthesis was designed in a multi-step process, including image processing and segmentation, mesh processing, and implant components design.

Image Processing and Segmentation

The diagnostic DiCOM images of the patient were processed using the ITK-SNAP software. Images were imported into a new workspace, after which semi-automatic segmentation was conducted. The image contrast was subsequently optimized for the cortical bone tissue, typically found in the Hounsfield Number range 200 – 2000. The image contrast was adjusted accordingly to focus on the bone tissue, relegating other anatomical structures out of focus. Subsequently, the snake tool was initialized, after which the region of interest was specified and semi-automatic segmentation was conducted using the thresholding algorithm, which expands and shrinks seed-bubbles to fit within defined contrast thresholds, using attractive and repulsive forces that equilibrate at threshold boundaries.

Following this procedure, the hip region and femurs were segmented separately to enable focus on the various components of the hip prosthesis. The segments obtained as STL files were subsequently polished using the meshmixer software to properly clean up the 3D segmented images and obtain smoother models that can be used for implant design. A screensnip of the raw hip region segment is presented in Figure 1, while the polished mesh files, imported into the Rhino workspace are shown in Figure 3.

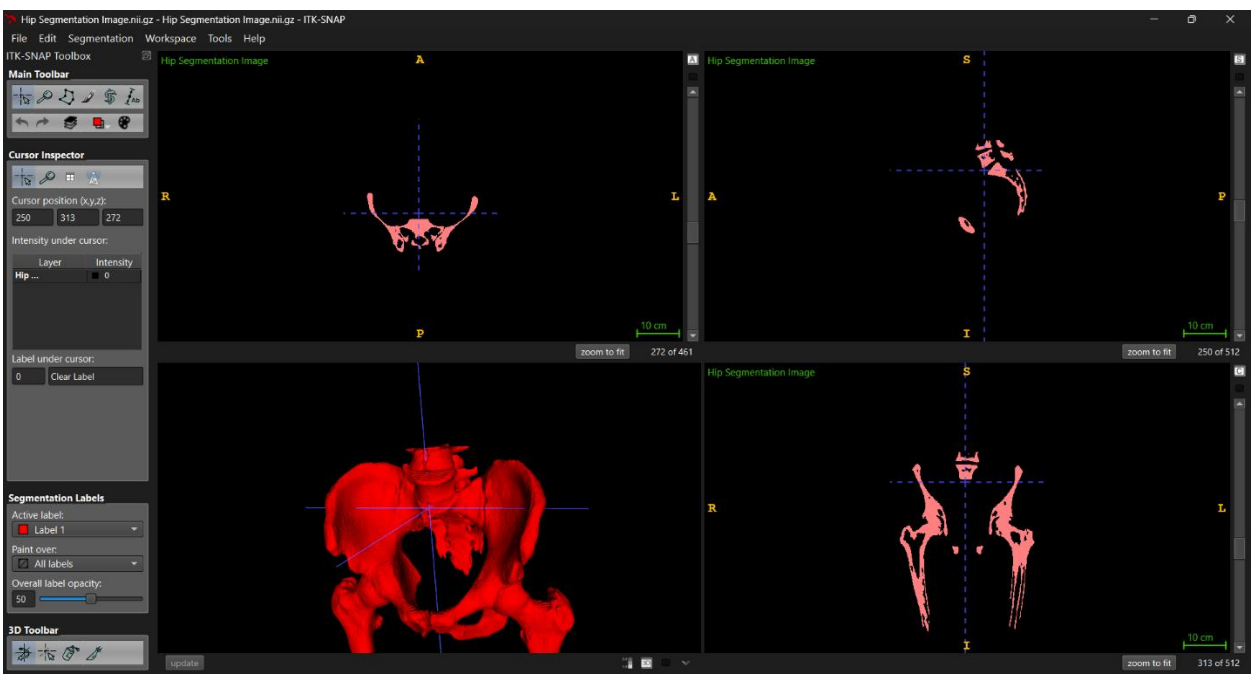


Figure 1. The raw hip and femur segments obtained by from the provided patient's DICOM images

Implant Components Design

The hip prosthesis comprises a femoral stem, femoral head, and acetabular construct as illustrated in Figure 2.

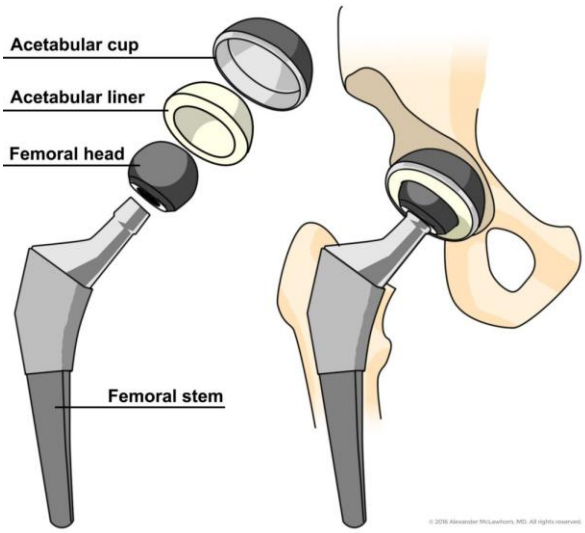


Figure 2. Components of the hip implant

In order to design each of these components to anatomically fit the patient, and incorporate biomechanical considerations reported in literature to improve performance and reduce the risk of failure, the Rhinoceros 3D software was used. First, the polished, segmented mesh files of the patient's hip and femoral bone tissues were imported into a rhino workspace. Both segments were superimposed and aligned accordingly as shown in the screen-clip provided in Figure 3.

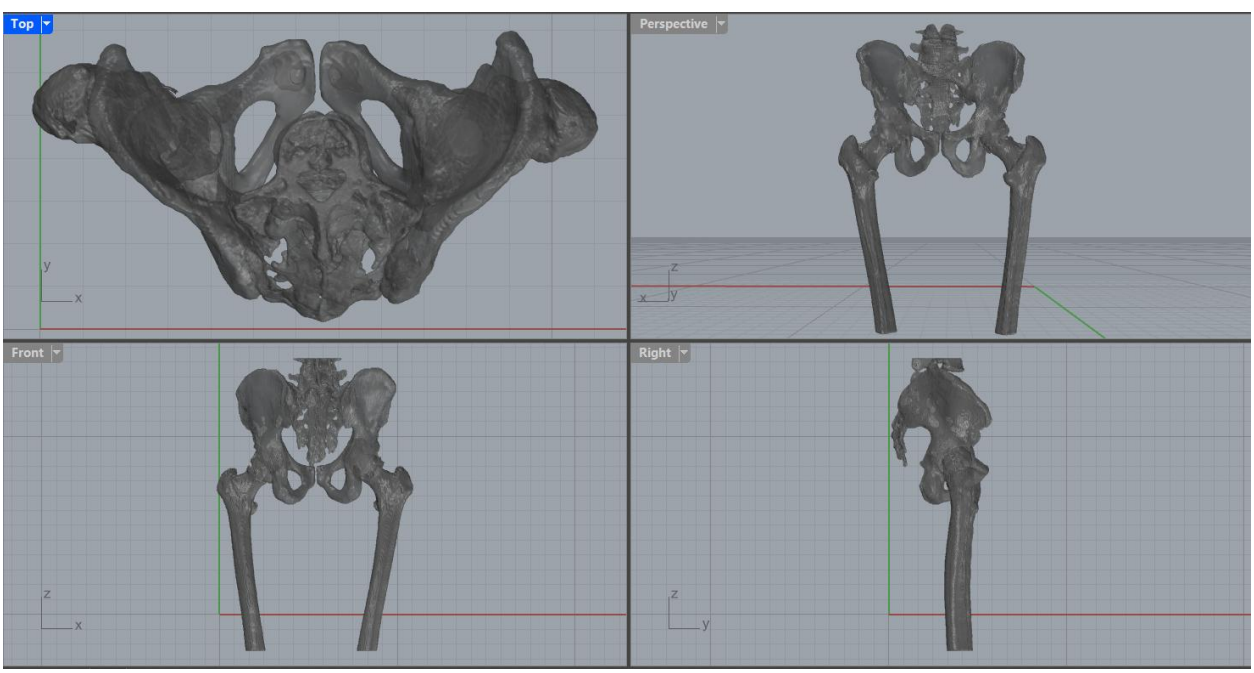


Figure 3. The superimposed and aligned hip and femoral bone tissue segments

Having imported the segmented images and aligned them accordingly, each of the prosthetic component was designed following the methods outlined below.

The Acetabulum and Femoral Head:

The acetabular cup and femoral head components of the implant were designed following simple constructive solids geometry (CSG) operations. First, a sphere was created with diameter equal to the patient's femoral head. Subsequently, a cylinder was created from the center of the sphere with a length roughly aligning with the original femoral neck from the patient's image. Rotations were conducted in all planes to ensure alignment of the cylinder with the patient's femoral neck. Subsequently, a Boolean difference operation was conducted to eliminate the intersecting volume of the femoral neck from the spherical femoral head. Next, the femoral head was offset in the outside direction to generate an acetabular cup, after which a plane was created to split the acetabular cup into a semi-sphere cup shape as required. The acetabular cup was capped and another Boolean difference operation was conducted to subtract the femoral head volume from the acetabular cup to ensure snug fit. A screen-snip of the acetabular cup obtained is shown in Figure 4, while the fit of the femoral head to the acetabular cup/socket is shown in Figure 5.

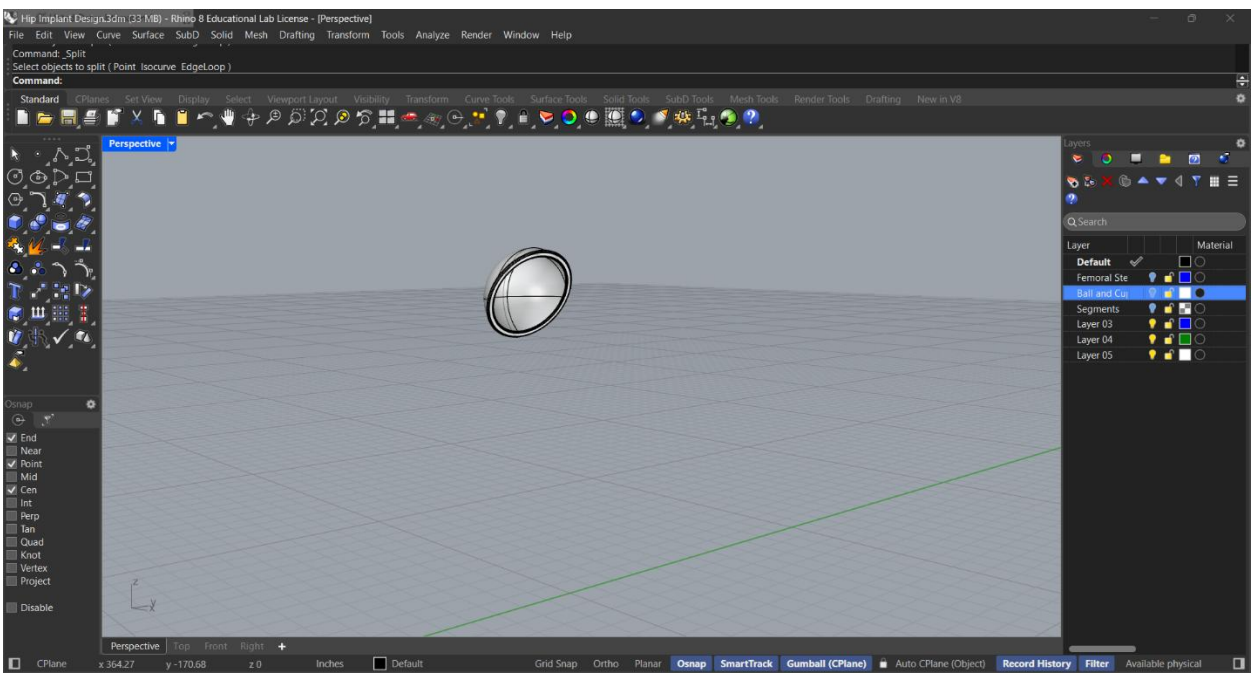


Figure 4. The designed acetabular cup

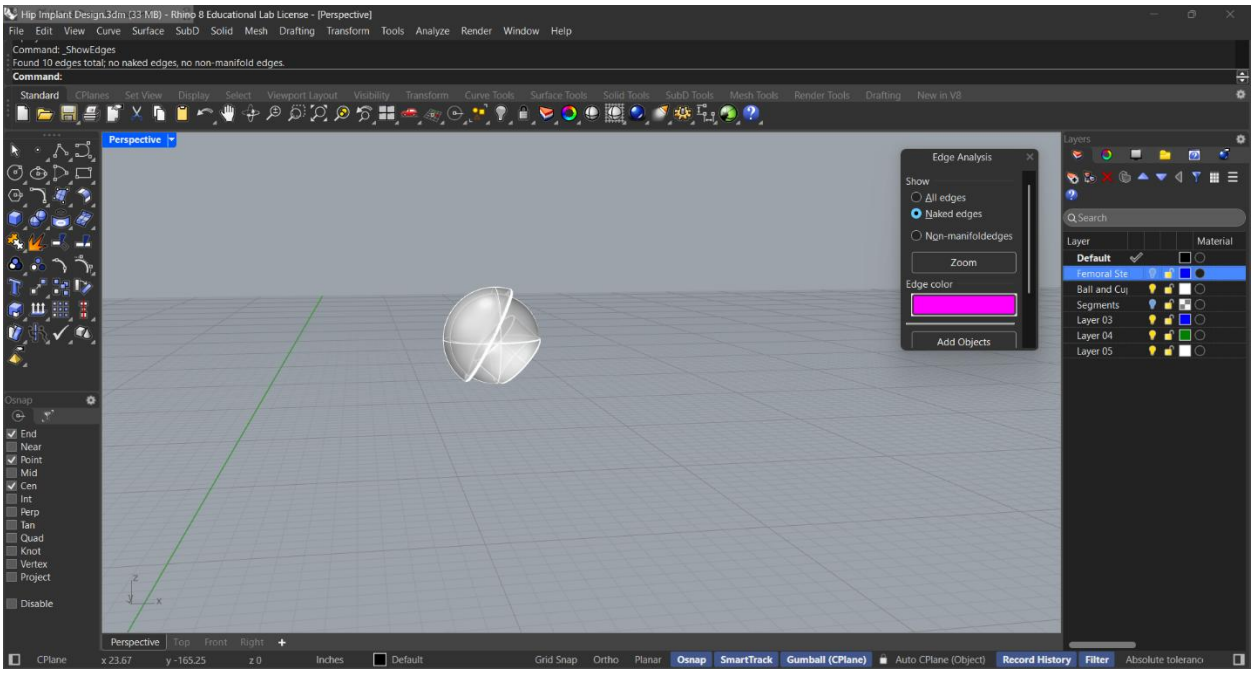


Figure 5. The fit of the femoral head to the acetabular cup

Femoral Neck and Stem:

The femoral component is about the most crucial part of the hip prosthesis, given the objectives of the project to ensure customization to the patient's anatomy and design optimizations. Hence, contrary to prototype methods described in literature, the femoral stem was designed using the patient's own femur as a template. To achieve this, cross sections of the patient's femoral bone were obtained along the transverse axis, using the rhino "section" command. Before obtaining the cross sections, the femur was profiled. A desired length (d) of the femoral stem was determined (130 mm) and a line segment was inserted at position z = 0, to intersect with the center of the femoral segment, considering the femoral head as position z = 130 mm. Subsequently, planes were placed at positions 0, 10, 30, 70, and 130 as illustrated in Figure 6. Contours were obtained using these line segments, since a smoother geometry is required to prevent uneven load distribution and risk of fracture that arises from jagged or uneven surfaces that would otherwise be obtained if all cross-sections of the patient's femur were used, while still maintaining a key surface profile that fits the patient's femoral inner contour.

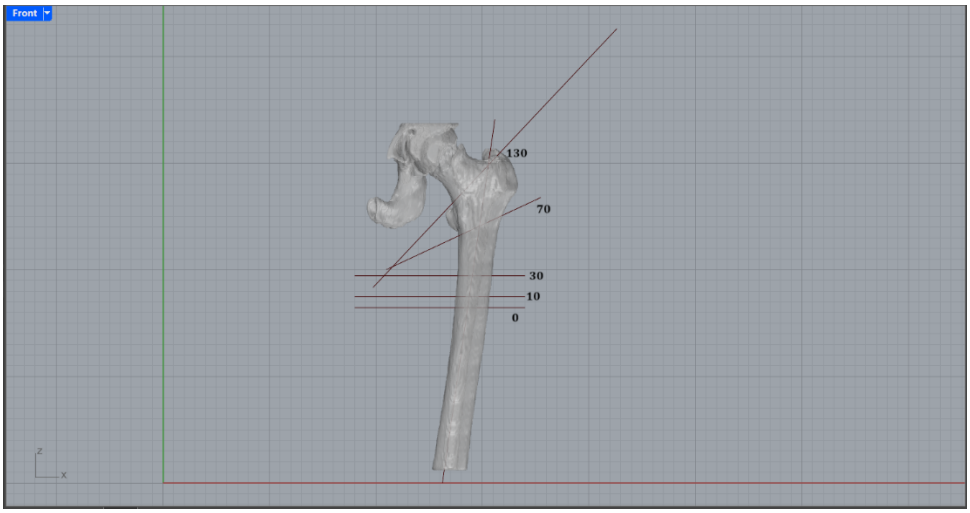


Figure 6. x-y Planes indicating the positions at which femoral cross sections will be used, and a line through the transverse axis of the femoral segment.

Having obtained the cross sections, all outer contours were deleted as only inner contours are required, except at position 130 where the stem will not be inserted into the femur (at 130, the outer contour was used). Additionally, slight changes were made to the contours through removal of some control points to achieve smoother curvatures and slight translation of some control points to provide more elliptical cross sections especially at proximal parts of the femoral stem to improve stress distribution without significantly compromising surface area of contact for osseointegration.[7] The most proximal contours (130 and 70) were adjusted using the rebuild command to 15-control point curves with a degree of freedom 3, while the rest were rebuilt to 10-control point curves, all with degrees of freedom 3. Subsequently, some control points were manually deleted and others slightly moved/adjusted to obtain the smoother cross-sections as illustrated in Figure 7.

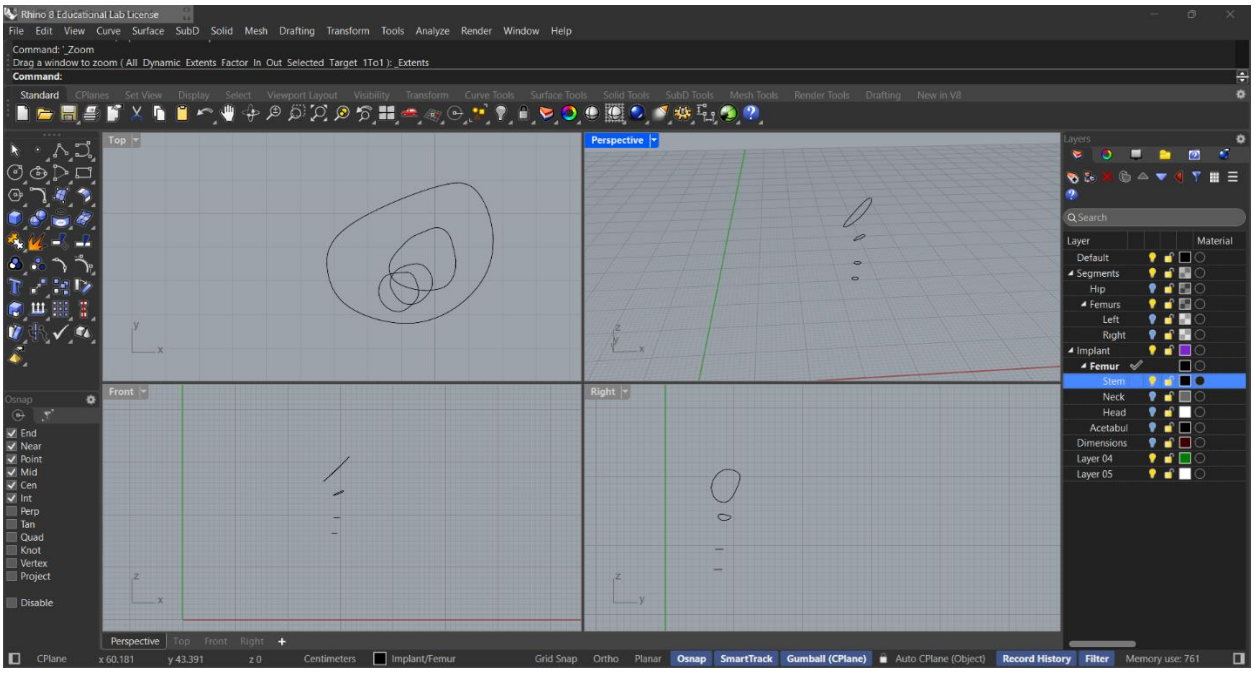


Figure 7. Selected cross-sections obtained from the patient's femur

Having smoothened and slightly re-shaped the contours, the femoral stem was generated by conducting a “loft” operation through all the contours to obtain the femoral stem implant shown in Figure 8.

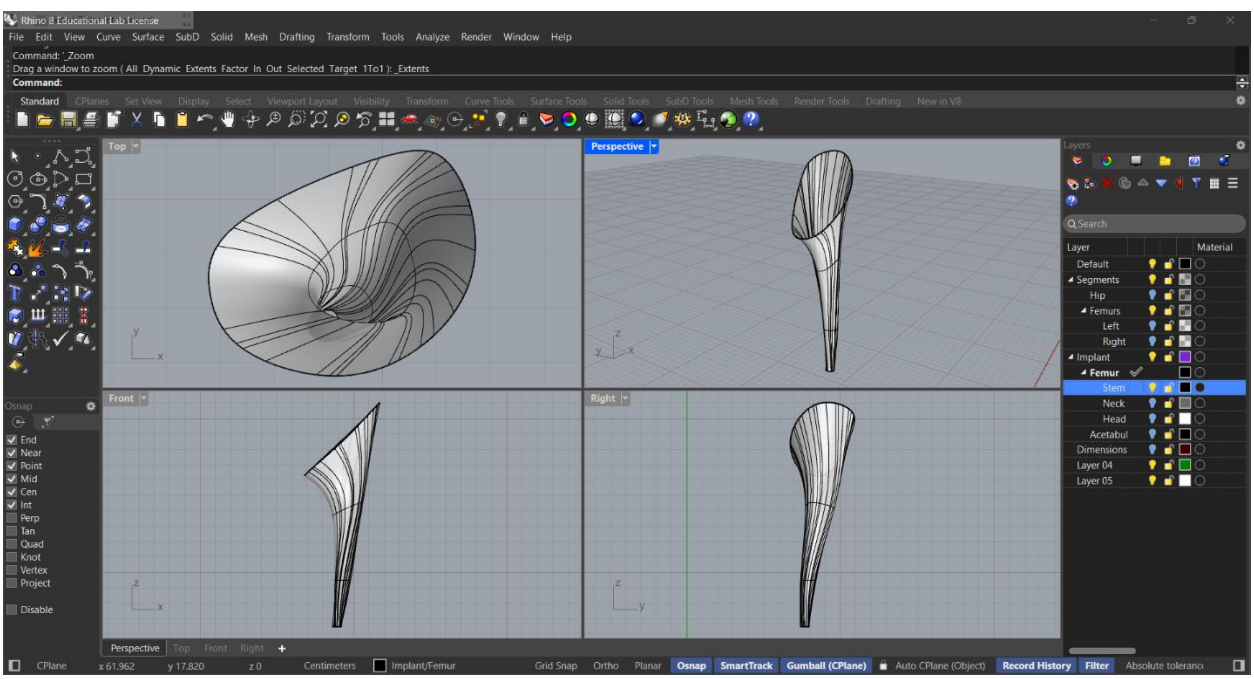


Figure 8. The femoral stem implant obtained as a surface from lofting the smoothened contours

After obtaining the BREP surface of the femoral stem, a blend surface command was done to join both the stem and femoral head surfaces, visually optimizing the seam and curvature of the adjoining surface for roundness and stress distribution. The full femoral implant BREP obtained is illustrated in Figure 9.

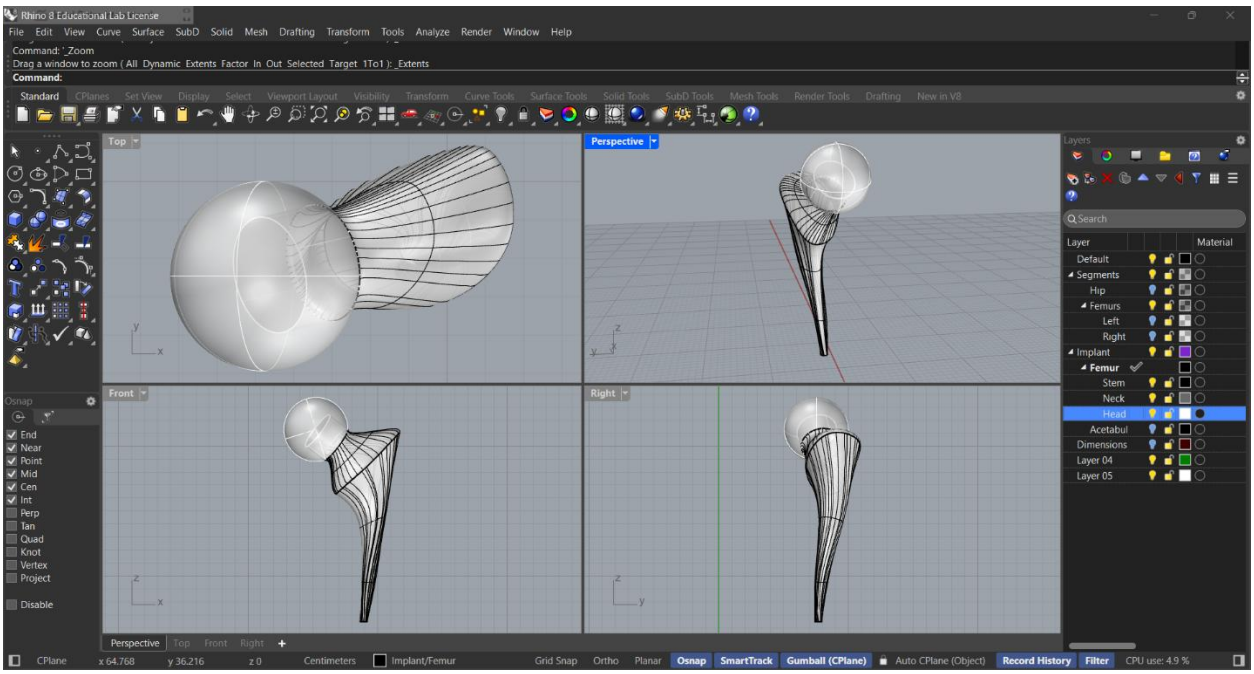


Figure 9. Full femoral implant

Applications of Wolff's bone remodeling law identify femoral regions where bone density is typically higher in healthy patients, and where implants fail due to insufficient osseointegration or implant mechanical failure.[4,8] To forestall these effects, first, two elliptical support structures were inserted into the femoral stem on both sides, just inferior to the femoral neck. The final femoral step BREP object obtained is illustrated in Figure 10.

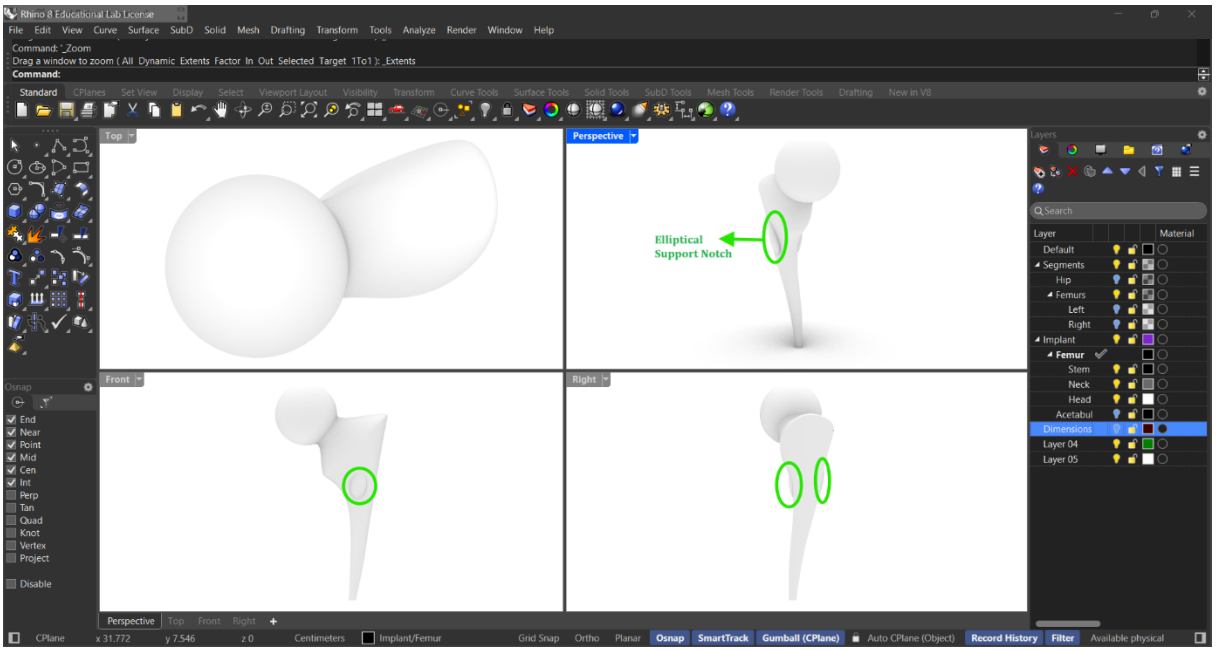


Figure 10. The final femoral stem implant BREP showing the elliptical support notch structures to improve stress distribution and osseointegration specifically at sites prone to failure and bone resorption

Subsequently, a rhinoscript code was generated to implement a porosity gradient across the entirety of the femoral stem, excluding the neck, since the femoral stem which will be inserted into the femur will be required to support cellular ingrowth and prevent stress-shielding. While it is beyond the scope of the current project to optimize lattice structures of the femoral stem implant, a porosity was nonetheless implemented across the transverse axis, from 70% proximal porosity to 40% porosity in the distal femoral regions, following literature reports.[9,10] An analysis of the rhinoscript code developed to implement this follows.

Rhinoscript code for porosity gradient

Step 1: Prompt user to select the BREP surface object, filament/fiber radius, and porosity bounds (minimum and maximum at distal and proximal regions respectively)

```
Call Main()
Sub Main()
    Dim i, j, strObject, arrBox, Line1, arrIntersectPoint, dblRadius, dblpor, dblpor1, dblpor2
    'Ask user to select an object and get its ID as strObject
    strObject = Rhino.GetObject("Select a solid object")
    'if there is no BREP object, exit
    If IsNull(strObject) Or (Rhino.IsSurface(strObject) = False And Rhino.IsPolysurface(strObject) = False) Then Exit Sub
    'Getting fiber radius and porosity values
    dblradius = Rhino.GetReal("enter the fiber radius", 0.02)
    dblpor1 = Rhino.GetReal("enter the proximal porosity in %", 40, 1, 99)
    dblpor2 = Rhino.GetReal("enter the distal porosity in %", 70, dblpor1, 99)
```

Step 2: Obtain the bounding box coordinates of the object and slice it into ‘intz’ number of 2D layers in the transverse axis, based on the filament/fiber diameter.

```
'get the 3D bounding box of the Surface
arrBox = Rhino.BoundingBox(strObject)
Dim OriginalBox: OriginalBox = arrBox
'Slicing the solid into intz 'Z-layers'
Dim intz, intx, inty, startpt, endpt
intz = 1 + (OriginalBox(4)(2) - OriginalBox(0)(2)) / (2 * dblradius)
```

Step 3: Initializing a for-loop iterating from slice z = 0 to intz. Then, implementing the linear porosity gradient from the lower to the upper bounds based on a linear function, which gives the porosity vs z-layer plot illustrated in Figure 11. Based on the calculated porosity at each layer, 1-dimensional x and y domains are calculated for each slice, essentially describing a 1D displacement comprising the filament/fiber diameter and air space, corresponding to the required porosity.

```
For i = 0 To intz
    Dim intdistz: intdistz = ((i - 1) * 2 * dblradius) + dblradius
    arrBox(0)(2) = OriginalBox(0)(2) + intdistz
    arrbox(1)(2) = OriginalBox(1)(2) + intdistz
    arrbox(3)(2) = OriginalBox(3)(2) + intdistz
    'Implementing the porosity gradient
    dblpor = dblpor1 + ((i / intz) * (dblpor2 - dblpor1))
    'Converting the porosity value (in %) to 1D interfiber spacing
    dblpor = (2 * dblradius * dblpor) / (100 - dblpor)
    'Dividing each Z-layer/slice inty 'Y-domains', and intx 'X-domains'
    intx = 1 + ((OriginalBox(1)(0) - OriginalBox(0)(0)) / ((2 * dblradius) + dblpor))
    inty = 1 + ((OriginalBox(3)(1) - OriginalBox(0)(1)) / ((2 * dblradius) + dblpor))
```

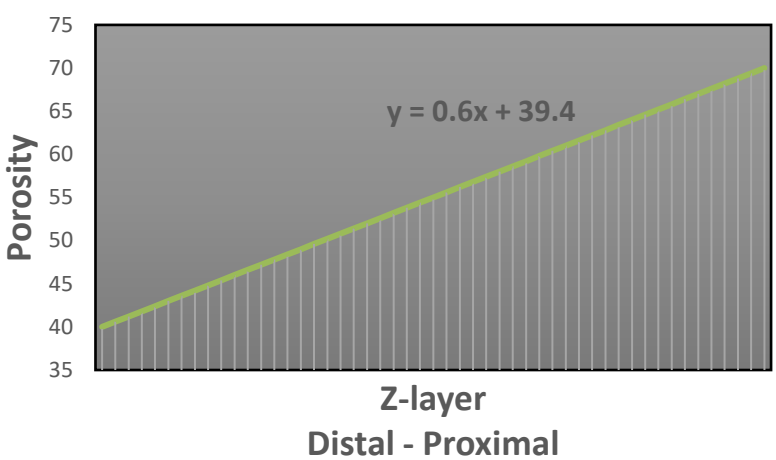


Figure 11. Plot of Porosity against z-layer demonstrating the linear porosity gradient demonstrated from 40% (distal) to 70% (proximal). The porosity bounds can be custom-set by a user in the user inputs as demonstrated in step 1.

Step 4: We alternate between y axis and x axis-oriented filaments/fibers to obtain a 0-90 lattice structure. For each orientation, we create a line at each 1D domain space as defined in step 3, calculate the intersection points of the line with the BREP surface, and enter a cylinder spanning the first and last intersection points (with radius = the user-defined fiber radius in step 1), excluding overlap or tangential intersections. Also, since we expect the femoral stem implant to be a solid volumetric BREP, we opt to not expend additional computing power, calculating for different contour intersection points in the cross-section to exclude some intersections that

preserve spaces on the inside of the shape. This is not required as the femoral stem implant has a solid shape without hollow spaces inside of it and it is therefore sufficient to simply calculate intersection points where the 1D lines enter and exit the shape to obtain representative porous structures of the solid.

Porosity Master code.rvb*

```
        If i Mod 2 = 1 Then
            For j = 0 To intx
                Dim intdistx: intdistx = (((j - 1) * dblradius) + ((j - 1) * dblpor) + dblradius)
                arrBox(0)(0) = OriginalBox(0)(0) + intdistx
                arrBox(0)(1) = OriginalBox(0)(1)
                arrBox(3)(0) = OriginalBox(3)(0) + intdistx
                Line1 = Rhino.AddLine(arrBox(0), arrBox(3))
                arrIntersectPoint = Rhino.CurveSurfaceIntersection(Line1, strObject)
                Rhino.DeleteObject Line1
                If IsArray(arrIntersectPoint) Then 'Excluding non-intersects
                    If arrIntersectPoint(0, 0) = 1 Then 'Checking to make sure intersection is a point intersection
                        startpt = arrIntersectPoint(0, 1)
                        endpt = arrIntersectPoint(Ubound(arrIntersectPoint, 1), 1)
                        Call Rhino.AddCylinder(startpt, endpt, dblRadius)
                    End If
                End If
            Next
        Else
            For j = 0 To inty + dblradius
                Dim intdisty: intdisty = (((j - 1) * dblradius) + ((j - 1) * dblpor) + dblradius)
                arrBox(0)(1) = OriginalBox(0)(1) + intdisty
                arrBox(0)(0) = OriginalBox(0)(0)
                arrBox(1)(1) = OriginalBox(1)(1) + intdisty
                Line1 = Rhino.AddLine(arrBox(0), arrBox(1))
                arrIntersectPoint = Rhino.CurveSurfaceIntersection(Line1, strObject)
                Rhino.DeleteObject Line1
                If IsArray(arrIntersectPoint) Then
                    If arrIntersectPoint(0, 0) = 1 Then
                        startpt = arrIntersectPoint(0, 1)
                        endpt = arrIntersectPoint(Ubound(arrIntersectPoint, 1), 1)
                        Call Rhino.AddCylinder(startpt, endpt, dblRadius)
                    End If
                End If
            Next
        End If

    Next
    'Rhino.DeleteObject StrObject

End Sub
```

Step 5: After implementing the gradient porosity for the femoral stem, a uniform porosity (50%) was implemented for the elliptical support notches (Figure 10), to provide mechanical rigidity, stress distribution within the implant, and provide higher surface area for osseointegration. Subsequently, two splitting planes were generated 0.5 inch from the distal end of the implant stem and 1 inch from the proximal edge, to split the stem and delete the middle surface section, while preserving and capping the terminal sections to assemble the femoral implant into one solid printable unit as illustrated in Figure 12.

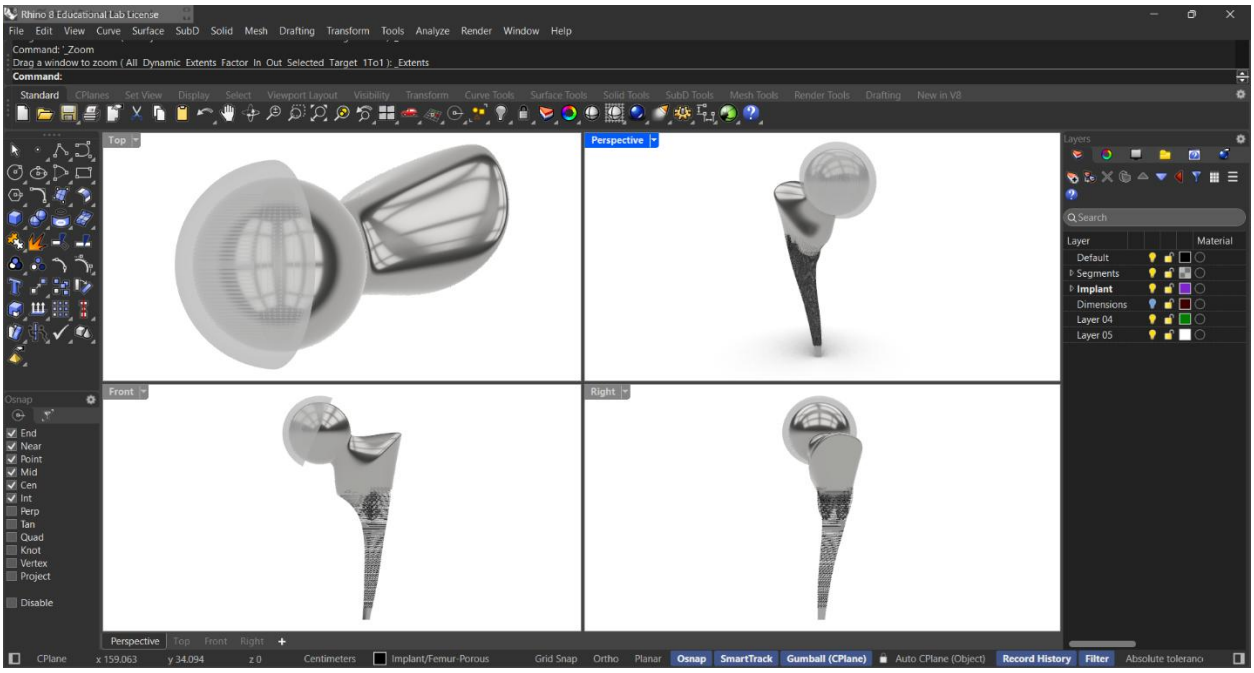


Figure 12. Final assembled hip prosthesis

Results and Analysis

The Final prosthesis obtained and fit with the patient's bone tissue segment are presented in Figure 13.

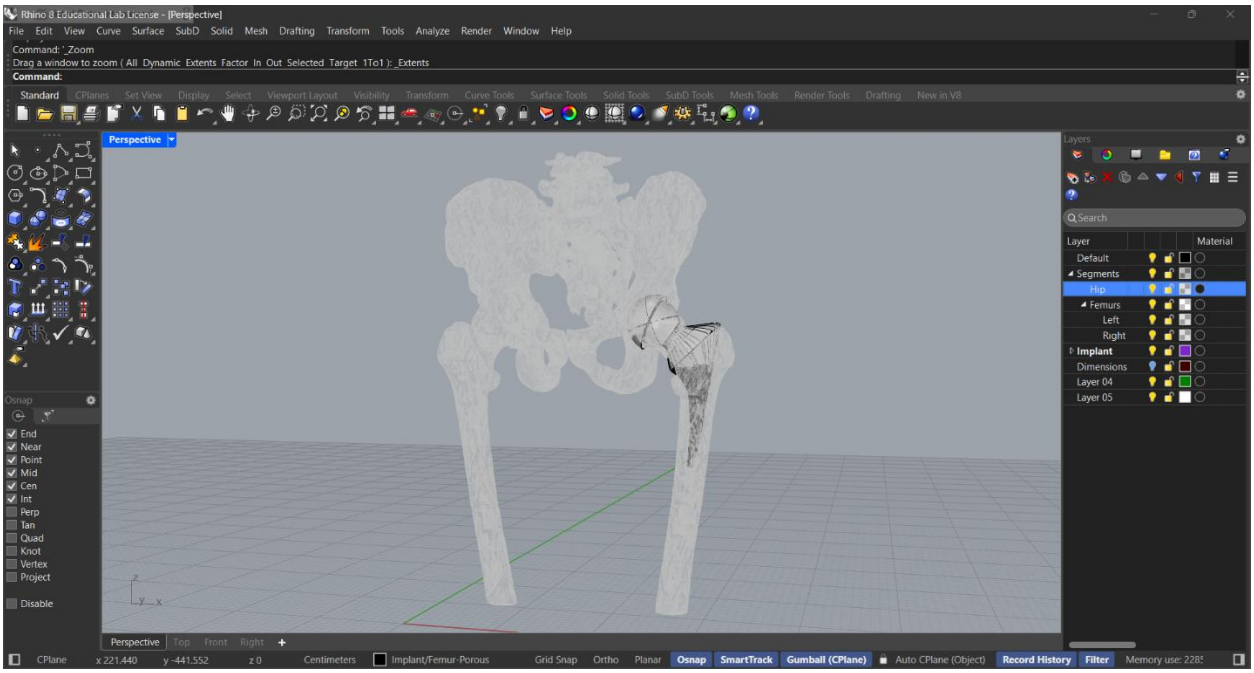
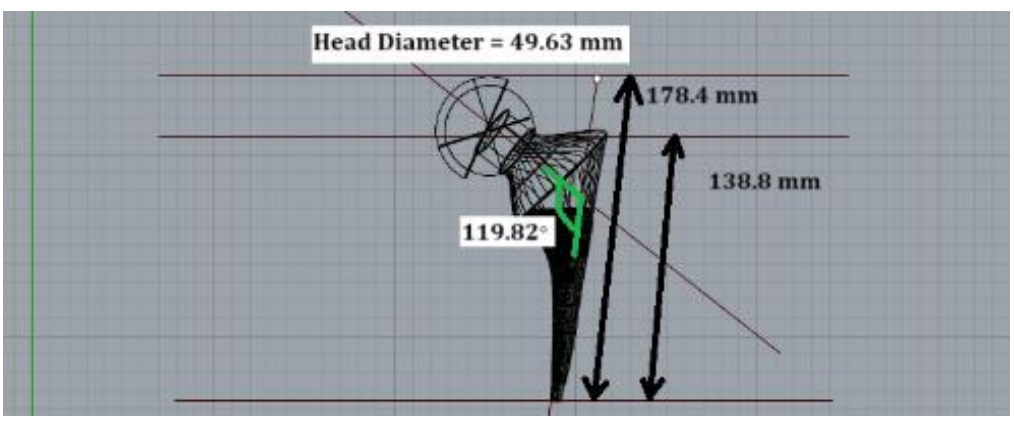


Figure 13. Fit of fully assembled prosthesis with patient's bone segment. Filament diameter = 760 µm

Structural and dimensional features of the implant are outlined as follows.



<u>Structural and Biomechanical Outcomes</u>

The designed femoral stem implant successfully integrated patient-specific anatomical features with literature-driven optimizations for stress shielding reduction and osseointegration. Key results include:

Stem Length: 139 mm, balancing stability and proximal bone loading in accordance with patient's femoral anatomy, and literature reports of optimized stem length 100 – 130mm.[11,12]
Stem Neck: Short, rounded neck surface with high thickness close to stem junction and about 120° angle promotes stress distribution and prevents early implant fracture after implantation.[12]
Cross-Sectional Geometry: Elliptical proximal sections and trapezoidal-circular distal sections improves stress distribution compared to completely circular designs,[7] while grooves modelled after patient's femoral anatomy ensure good fit and integration with improved surface area for cellular attachment and osseointegration.
Porosity Gradient: Proximal Region with 70% porosity to mimic trabecular bone stiffness (~2.5 GPa),[9] and distal Region with 40% porosity for load bearing.[10]
Elliptical Support Notches: 50% uniform porosity enhanced stress transfer to high-density femoral regions.[8]

Additive Manufacturing Feasibility

The design complied with SLM constraints of strut Diameter: 150 µm fiber size (above SLM's 100 µm minimum feature size). Additionally, no naked edges or manifold edges exist in the implant model as illustrated in Figure 14.

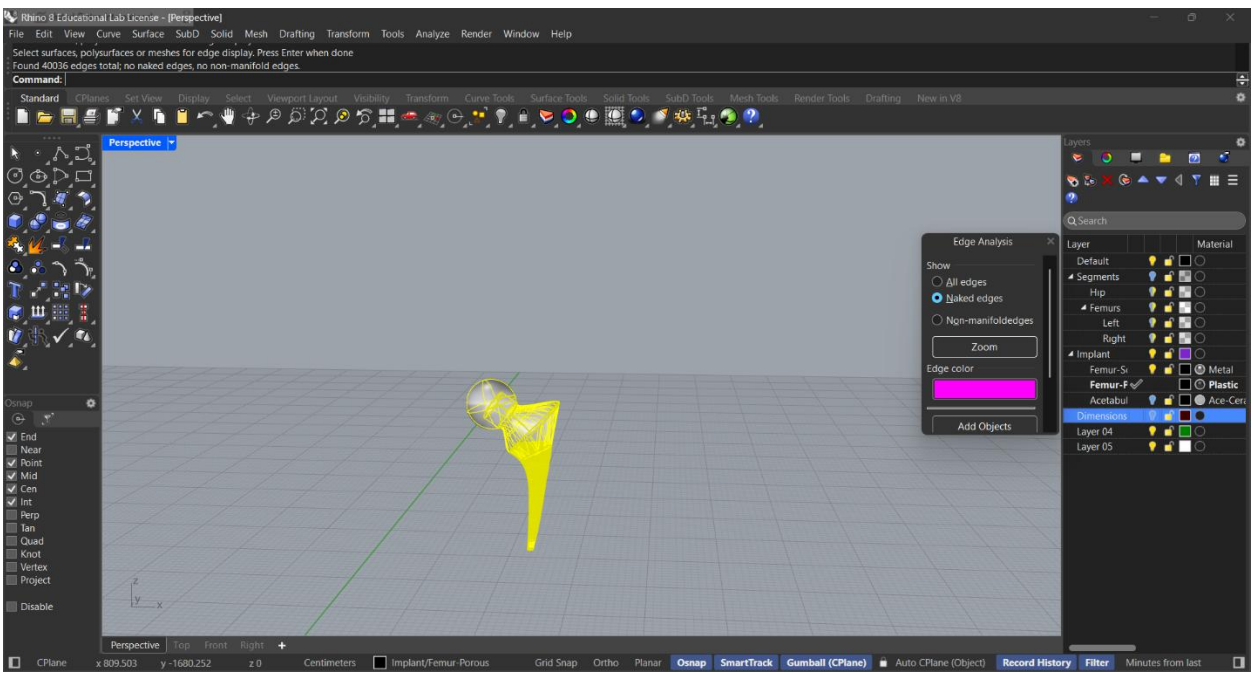


Figure 14. Edge analysis of the designed implant

Discussion and Conclusion

The hip prosthesis was successfully modelled and custom designed for the patient whose images were obtained and used. The graded porosity design (70% proximal – 40% distal) reduces stiffness mismatch.[13] The rhinoscript code generated also allowed for user-input porosity gradient to adjust and optimize the porosity gradient on a patient-by-patient basis. The proximal porous lattice structure potentially lowers the effective modulus closer to cortical bone (17–20 GPa), while distal thicker struts prevent excessive micromotion and implant failure. This approach has been shown to outperform homogeneous porous stems, which exhibit 74% higher bone resorption.[13] The inclined elliptical notches (Figure 10) also promotes directional bone growth, and enhanced ingrowth in high stress areas of the femur.

There are limitations in the scope of the current project and corresponding indications for future work. Appropriate computational studies, including finite element analyses, are required to properly characterize stress-shielding, micromotion, and Von Mises stress features of the implant. Additionally, future research should prioritize fatigue testing since long-term cyclic loading tests are needed to validate the durability of the femoral stems, beyond FEA predictions. Lattice optimizations like TPMS structures (e.g., Gyroid) may further enhance strength-to-weight ratios as demonstrated by Kladovasilakis et al.[9] Material optimization should also be considered, with various material alternatives explored to improve biocompatibility and performance.[14]

In conclusion, this project developed a patient-specific femoral stem implant combining anatomically accurate geometry with lofted contours from segmented STL data, ensuring anatomical fit. Biomechanical optimization was also attempted, with graded porosity and elliptical notches to reduce stress shielding. The designed implant is manufacturing ready, without naked or manifold edges and printable filament/fiber diameter. However, thorough fatigue testing is recommended to better validate the applicability of the design and approach. Nonetheless, the methodology described can be extended to other joint replacement prostheses using similar porosity-gradient and patient anatomy–specific strategies.

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